# Mathematical Modelling

(see References)

# So last meeting, we learned about

- applications of M/M/1 in which the utilization factor is computed
- Use MS Excel in setting up a simulation model applying Discrete-Event Scheduling in M/M/1

# Mathematical Modelling

**Mathematical modeling** is the process of translating real-world problems into mathematical languages, like algorithms, to understand, analyze, and predict behavior.

It serves as the **bridge between abstract theory and practical application.**

# The Modelling Process

The modelling process is commonly consisting of 5 phases.

1

**Define the specific question**

Ask questions

2

**Make assumptions**

Simply the real-world and make it mathematically manageable

3

**Construct the model**

Build mathematical equations

4

**Solve the problem**

Use the mathematical eq. to find a solution

5

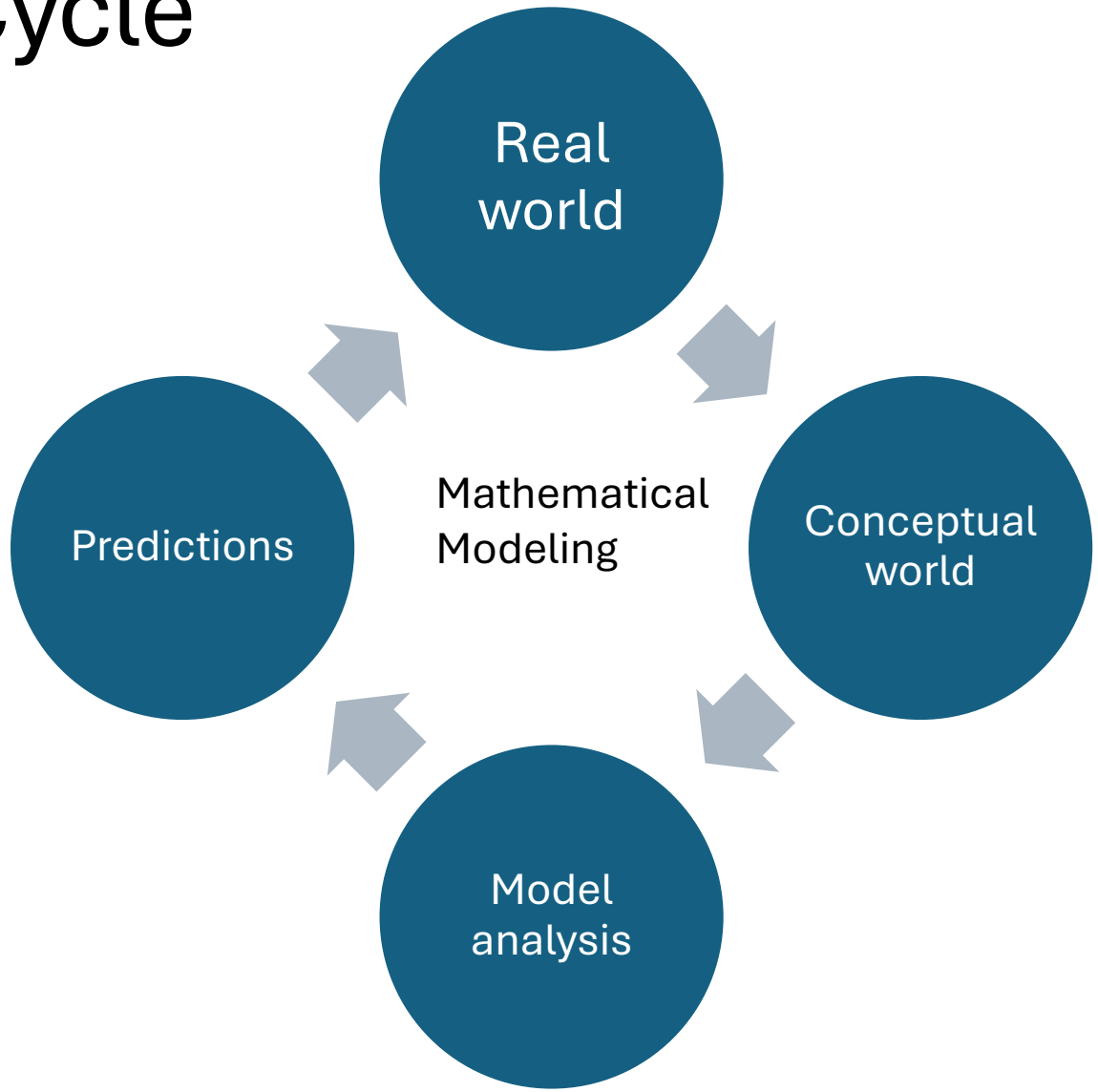
**Validate/ Evaluate**

Compare the mathematical results against the real-world observations

# Mathematical Languages

- Algebraic equations
- Differential equations
- Integral equations
- Algorithms
- Formulas
- Theorems

# The Modeling Cycle



# Monte Carlo Simulation

- A mathematical technique that uses random sampling to forecast (predict – to speak about future events).
- It is done in executing a model many times with random inputs while it builds a spectrum of likely results.

# Example 1

A shop sells product A. Past data show that daily demand varies from 0 to 3 with probabilities of 10%, 30%, 40%, and 20%, respectively. The owner wants to simulate demand for 5 days to estimate the average daily sales. The simulation trials would use 45, 12, 82, 37, and 65.

# Probability distribution of demand

| # of units sold | Probability | Cumulative Probability | Random Digit Assignment (Random Number Interval, 00 – 99) |
|---|---|---|---|
| 0 | 0.10 | 0.10 | 00 – 09 |
| 1 | 0.30 | 0.40 | 10 – 39 |
| 2 | 0.40 | 0.80 | 40 – 79 |
| 3 | 0.20 | 1.00 | 80 - 99 |

## Simulation Trials

| | |
|---|---|
| Day 1 | 45 |
| Day 2 | 12 |
| Day 3 | 82 |
| Day 4 | 37 |
| Day 5 | 65 |

## Solutions and Results

| | | | |
|---|---|---|---|
| Day 1 | 45 | Falls in 40– 79 | 2 units |
| Day 2 | 12 | Falls in 10 – 39 | 1 unit |
| Day 3 | 82 | Falls in 80 – 99 | 3 units |
| Day 4 | 37 | Falls in 10 – 39 | 1 unit |
| Day 5 | 65 | Falls in 40 - 79 | 2 units |

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## Answer

**Total simulated demand over 5 days**
**= 2 + 1 + 3 + 1 + 2 = 9 units**

**Average daily demand**
**= 9/5 = 1.8 units/day**

# Example 2

A repair shop fixes laptops in 2,3, or 4 hours with the following probabilities:

2 hours (30% chance);

3 hours (50% chance); and

4 hours (20% chance).

Simulate a 5 repair jobs using random numbers 45,12, 85, 67 and 29, to find the average repair time.

# Probability distribution of t-shirt printing

| # of hours to fix | Probability | Cumulative Probability | Random Digit Assignment (Random Number Interval, 00 – 99) |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

## Simulation Trials

| | |
|---|---|
| Day 1 | 45 |
| Day 2 | 12 |
| Day 3 | 85 |
| Day 4 | 67 |
| Day 5 | 29 |

## Solutions and Results

| | | | |
|---|---|---|---|
| Day 1 | 45 | | |
| Day 2 | 12 | | |
| Day 3 | 85 | | |
| Day 4 | 67 | | |
| Day 5 | 29 | | |

**Total simulated repairs over 5 days = ?**

**Average repair time. = ?**

# Example 2a

A repair shop fixes laptops in 1, 2, 3, or 4 hours with equal probabilities. Simulate a 5-day repair jobs using random numbers 45,12, 85, 67 and 29, to find the average repair time.

# Example 3

A t-shirt printing machine prints 15, 20, 25, 30 t-shirts with the following probabilities:

15 (20% chance);
20 (35% chance);
25 (25% chance); and
30 (20% chance).

Simulate a 7-day printing using random numbers 45, 34, 40, 25, 28, 50, 15, respectively. Find the average number of t-shirts printed.

# Probability distribution of t-shirt printing

| # of t-shirts printed | Probability | Cumulative Probability | Random Digit Assignment (Random Number Interval, 00 – 99) |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

## Simulation Trials

| | |
|---|---|
| Day 1 | |
| Day 2 | |
| Day 3 | |
| Day 4 | |
| Day 5 | |
| Day 6 | |
| Day 7 | |

## Solutions and Results

| | | | |
|---|---|---|---|
| Day 1 | | | |
| Day 2 | | | |
| Day 3 | | | |
| Day 4 | | | |
| Day 5 | | | |
| Day 6 | | | |
| Day 7 | | | |

**Total simulated printing over 7 days = ?**

**Average daily printing = ?**

# Stochastic Modeling

It is mathematical method that uses **probability distributions** to estimate potential outcomes by **factoring in random variations** over time.

It simulates systems to **generate a range of probable results** along with their **likelihood.**

# Basic Concepts underlying Stochastic Modeling

**Random Variables** : values that change based on probabilistic events rather than fixed rules

**Stochastic Processes** : sequence of indexed random variables

**Markov Chains** : systems where the next state depends only on the current state, not the past paths.

**Monte Carlo Simulation** : computational algorithm that rely on repeated random sampling to compute results

# Types of Monte Carlo

- Monte Carlo Simulation
- Markov Chain Monte Carlo
- Monte Carlo Tree Search
- Randomized Decision Algorithm

# Monte Carlo Tree Search



It is powerful in decision making and game AI by randomly sampling prospective **game moves** to build a search tree of optimal paths/future possibilities.



**Informed search**



Planning algo used for **AlphaGo**

# Four Steps of MCTS



SELECTION



EXPANSION



SIMULATION



BACK PROPAGATION

# Components of a Tree in MCTS

Circle : State

Solid Circle : Action

Line (with arrow): selected path (edges between nodes)

# MCTS

- Balances **exploration (trying new moves)** and **exploitation (deepening promising moves)** using **Upper Confidence Bound (UCB1)** score.

See formula for UCB1

# pseudocode

start
initialize current to 0
if current is a leaf-node then do
    if $n_i$ (current) is 0 then do
        rollout;
    else
        for each available action from current
            add a new state to the tree
        current is the selected (first) new child node
        rollout;
else
    current is the child node of the current that maximizes UCB1 ($S_i$)

# Thank you for listening!

# Reference(s)


1. “Monte Carlo Simulation”. Retrieved from: What Is Monte Carlo Simulation? | IBM or https://www.ibm.com/think/topics/monte-carlo-simulation
2. “Slide Design”. Generated in PPT Designer.